Lipids and Waxes

Lipids

- Physical properties
- Chemical properties
- Extraction methods
- Official fixed oils/fats and their composition
  - Arachis oil, castor oil, almond oil, sesame oil, theobroma oil, codliver oil etc

Waxes

- Definition
- General properties
- Animal waxes
- Vegetable waxes

Introduction

- Lipids are the substances of animal and plant origin and comprise of fixed oils, fats and waxes.
- The basic function of oils and fats is storage of energy, apart from their several uses in medicine and industries.
- Defined by their physical property and not by their structure.
- They are the fundamental building blocks of lipids
- *Most lipids are fatty acids or ester of fatty acids*

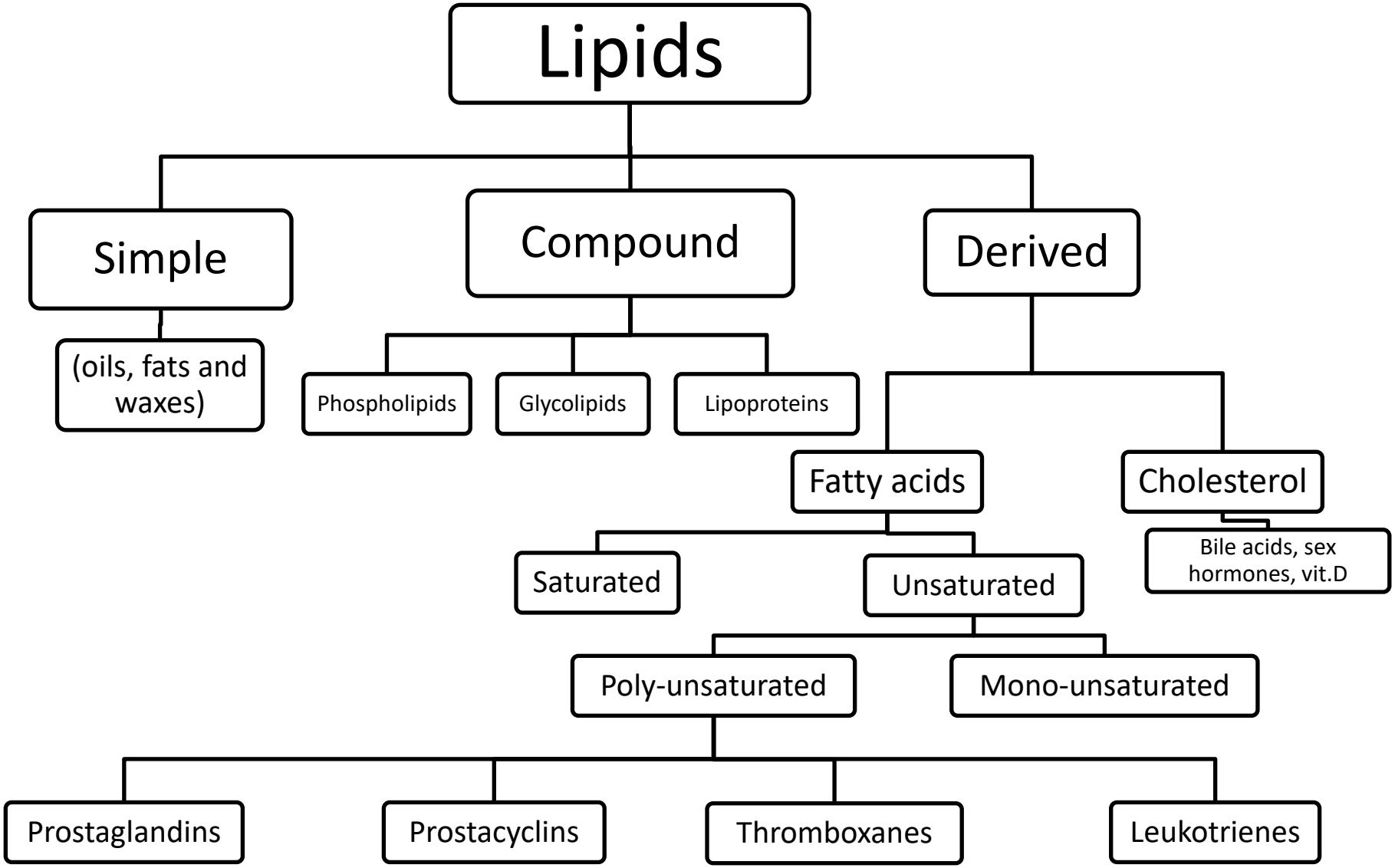
- They are soluble in non-polar solvents such as petroleum ether, benzene, chloroform
- They are carboxylic acids with hydrocarbon chains ranging from 4 to 36 carbons long.
- Those which are liquid at 15.5^0 to 16.5^0 are called fixed oils; which are solid or semi-solid at above temperature are termed as fats.
- Fixed-oils, derived from plant sources, occur generally from seeds.

- They possess the following properties.
  - Fixed oils are thick, yellow-coloured liquids with characteristic odour.
  - They are non-volatile and cannot be distilled.
  - They do have food value and can be saponified.
  - They turn rancid on storage due to free acidity.
  - Water insoluble compounds. Chains may be fully saturated, or have some unsaturated double bonds
  - Chains are usually unbranched

Definition

- Fats and oils are esters of glycerols and various straight chained monocarboxylic acids, known as fatty acids.
- The fatty acids of natural fats have 4 to 24 carbon atoms. These fatty acids may be saturated, monosaturated or polysaturated.
- A *fatty acid* is a molecule characterized by the presence of a carboxyl group attached to a long hydrocarbon chain.
- A *fatty acid* is a molecule with a formula R–COOH where R is a hydrocarbon chain
- Fatty acids can be said to be carboxylic acids

Classification of Lipids



Fat vs. Sugar

- Fats
  - Twice the energy per gram (since almost fully reduced) of sugars
  - Nearly insoluble in water
  - Energy release is slow (but just keeps going…)
- Sugars
  - Only half the energy/gm of fats
  - Very water soluble
  - Fast energy release (can't store more than 1 day)

Properties of fats and oils

- Fats are solids or semi solids
- Oils are liquids
- Melting points and boiling points are not usually sharp (most fats/oils are mixtures)
- Specific gravity < 1.
- When shaken with water, oils tend to emulsify
- Pure fats and oils are colorless and odorless (color and odor is always a result of contaminants) – i.e. butter (bacteria give flavor, carotene gives colour)

Properties of fats and oils

- Cannot be distilled (unless decomposed). This distinguishes fixed oils from volatile oils.
- Leaves a permanent, translucent stain on filter paper.
- All insoluble in water. Soluble in organic solvents. (Except ethyl alcohol [except castor oil]).
- Most develop a rancid odour when exposed to air, moisture & light for prolonged periods of time (hydrolysis of esters & liberation of fatty acids)
- All have characteristic odours.
- Varying viscosities.

Non-drying, semi-drying and drying oils

- based on the ease of autoxidation and polymerization of oils (important in paints and varnishes)
- the more unsaturation in the oil, the more likely the “drying” process
  - Non-drying oils:
    - Castor, olive, peanut, rapeseed oils
  - Semi-drying oils
    - Corn, sesame, cottonseed oils
  - Drying oils
    - Soybean, sunflower, hemp, linseed, tung, oiticica oil

Properties of fats and oils

i. Physical Properties
ii. Chemical Properties

PHYSICAL PROPERTIES

a. Specific Gravity
b. Melting / Congealing Point
c. Refractive Index
d. Viscosity
e. Optical Rotation

PHYSICAL PROPERTIES

SPECIFIC GRAVITY

- Fixed oils and fats all have a specific gravity of < 1.

- They are therefore all lighter than water.

MELTING / CONGEALING POINT

- This physical constant is only used for 2 oils (only 2 fixed oils are solid at room temperature).
  - Oil of Theobroma
  - Oil of Hydrocarpus

PHYSICAL PROPERTIES

REFRACTIVE INDEX

- **DEFINITION**: This is the ratio of the velocity of light in a vacuum to the velocity of light in a substance.
- Refractive index varies with the wavelength of light and temperature.
- Lab: wavelength is kept constant with using a Na-lamp, T=20 ºC. Refractive index is measured with a refractometer.
- The refractive index varies considerably for every oil and is expressed as a range

VISCOSITY

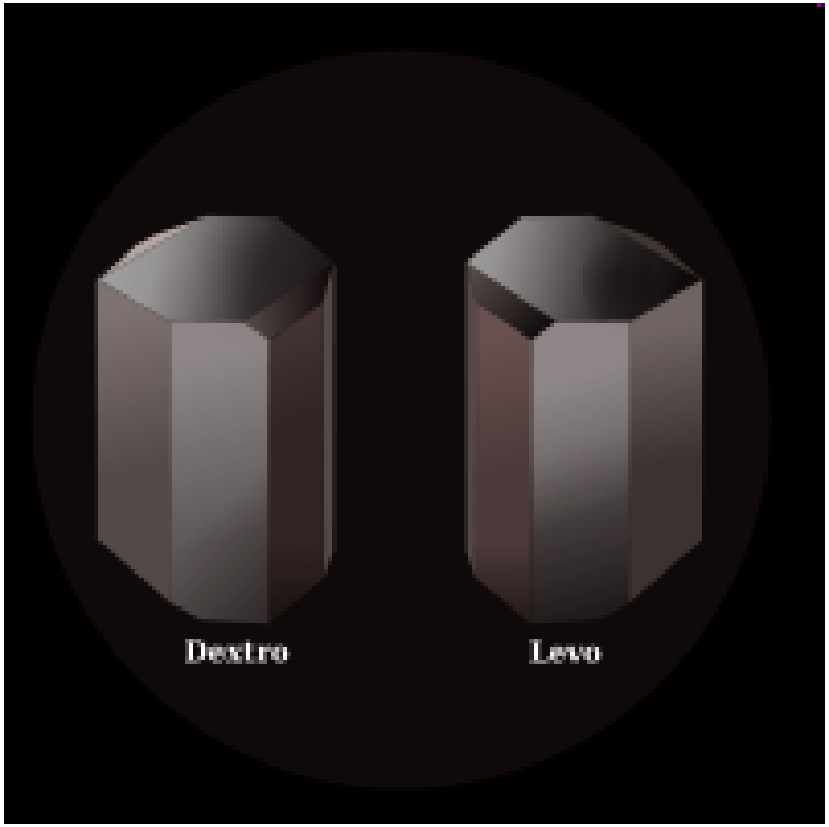
- Viscosity is the measure of a material's resistance to flow. Viscosity is a result of the internal friction of the material's molecules.
- The tendency of a fluid to resist internal flow without regard to its density.
- The resistance of fluid substance to flowing, quantitatively characteristic for an individual substance at a given temperature and under other definite external conditions.

PHYSICAL PROPERTIES

OPTICAL ROTATION / ACTIVITY

1. The angle through which the plane of polarization of light is rotated when the polarized light passes through a layer of liquid.

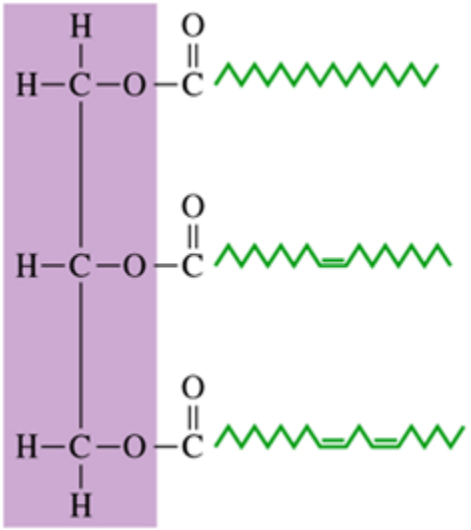
2. **Optical rotation** is the rotation of <u>linearly polarized light</u> as it travels through certain materials.



Chemical reactions of triacylglycerols

Reactions of triacylglycerols

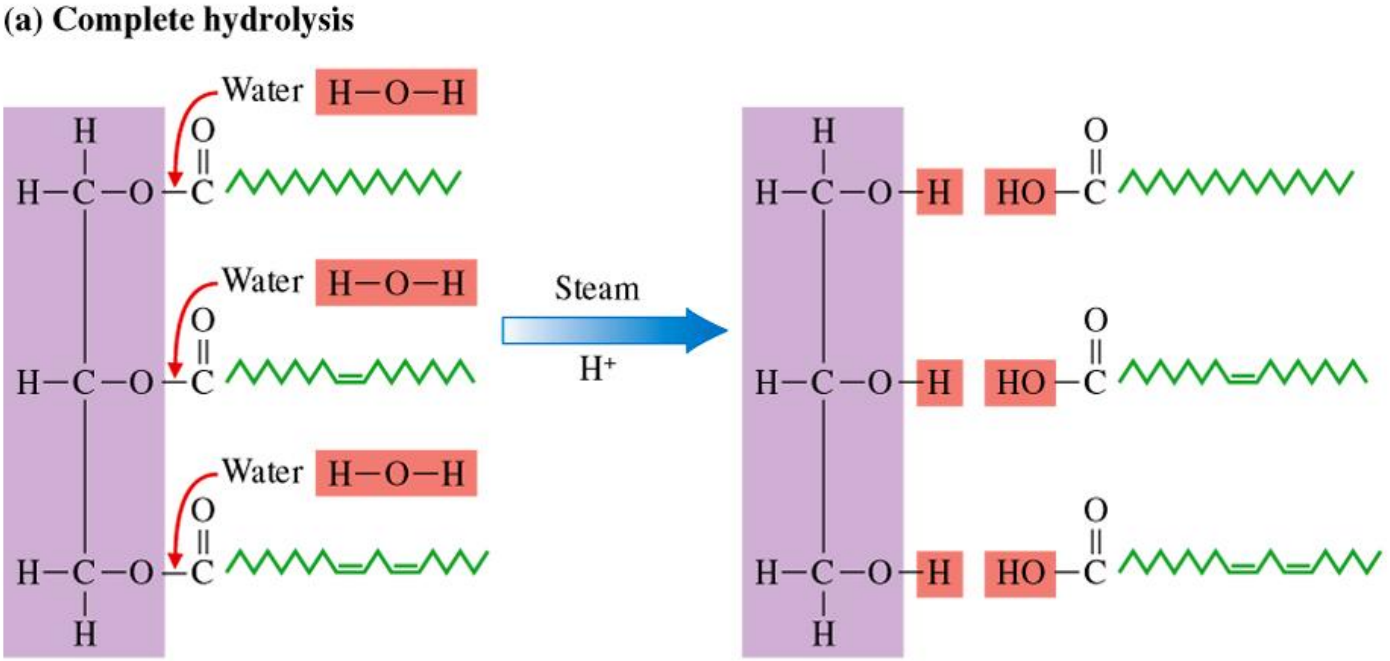
- Triacylglycerols have ester groups and (sometimes) double bonds, so the reactions that can happen with triacylglycerols involve those functional groups (hydrolysis, addition reactions)
- They can also undergo oxidation (enzyme-mediated) at the double bond to produce two carbonyl groups (aldehyde and then carboxylic acid).



Chemical reactions of triacylglycerols

Reactions of triacylglycerols

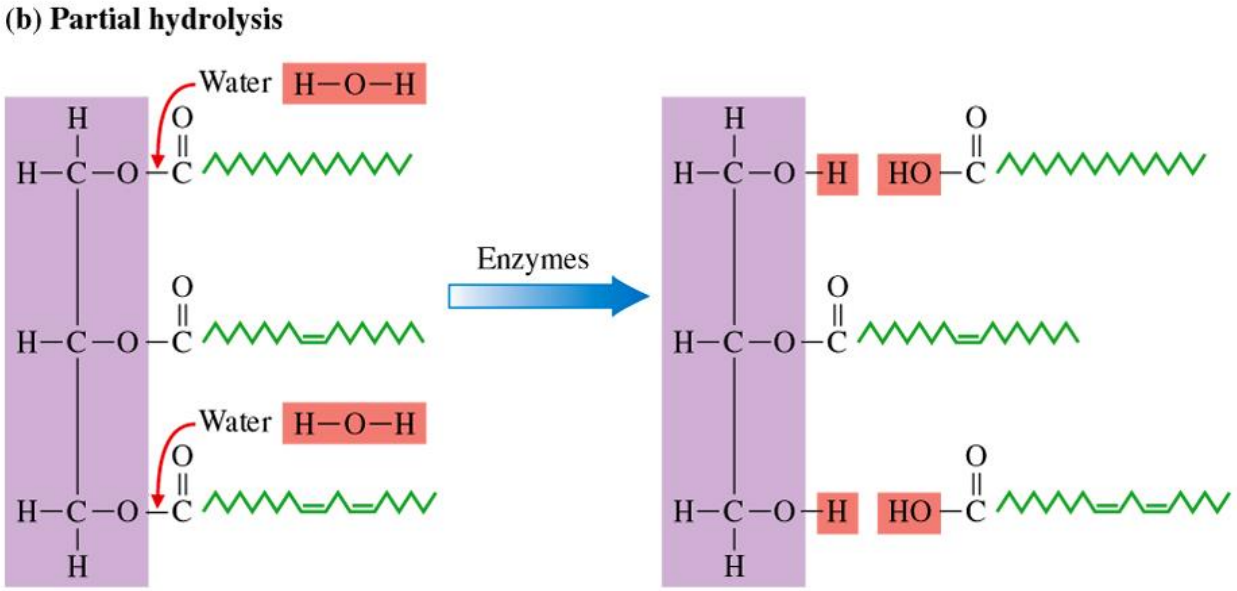
- **Hydrolysis**: since triacylglycerols are tri-esters, the ester groups can undergo hydrolysis to yield carboxylic acids and glycerol (1,2,3-Propanetriol).
- This kind of reaction is important in digestion.



Chemical reactions of triacylglycerols

Reactions of triacylglycerols

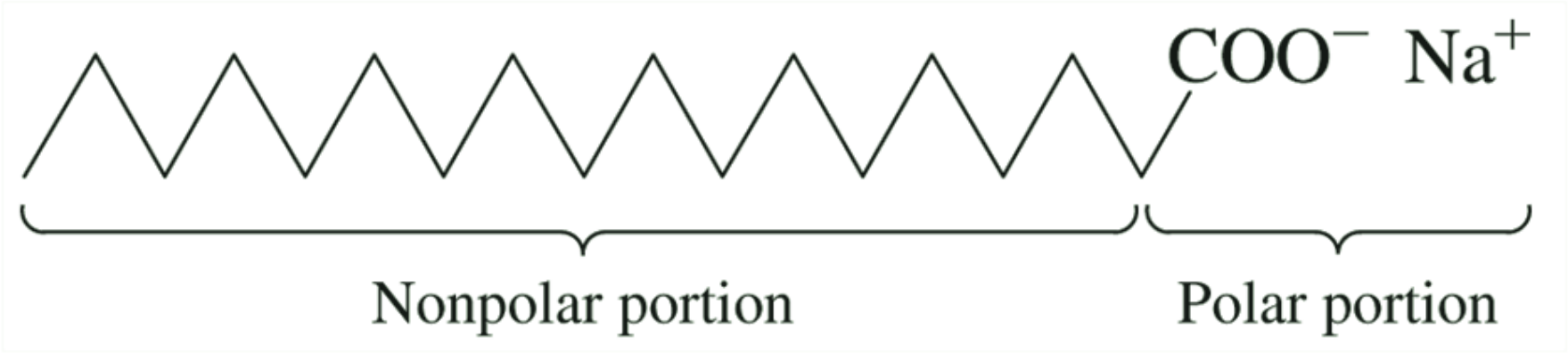
- Enzymes catalyze the hydrolysis of triacylglycerols in a stepwise fashion. The outside ester groups are hydrolyzed first before the middle group.
- In certain cases, triacylglycerols are only partially hydrolyzed (partial hydrolysis) – this happens when only one or two of the fatty acid groups are removed from the triacylglycerol by hydrolysis.



Chemical reactions of triacylglycerols

Reactions of triacylglycerols

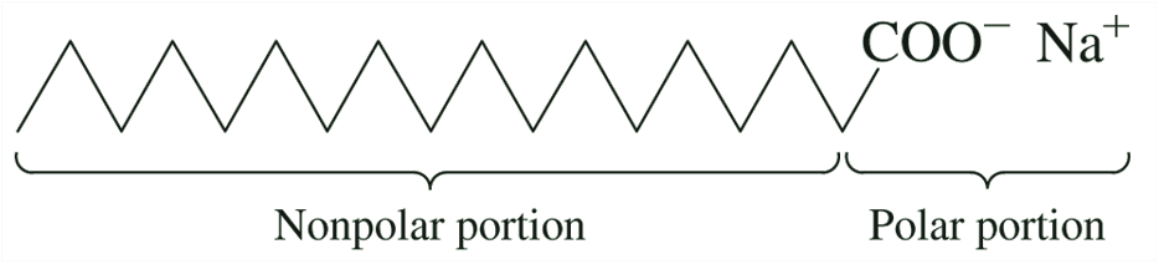
- **Saponification**: as before, the hydrolysis of an ester carried out under basic conditions is called saponification. The base causes de-protonation of the carboxylic acid to form a carboxylic acid salt (called fatty acid salts).

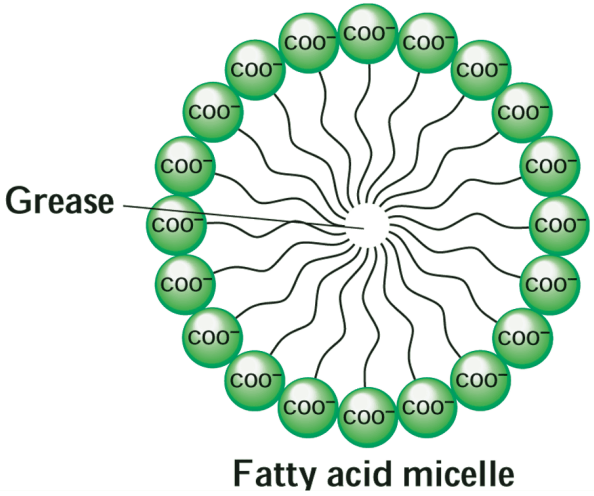


Chemical reactions of triacylglycerols

Reactions of triacylglycerols

- Saponification of a triacylglycerol yields fatty acid salts. The anions (negative ions) in these salts have long carbon chains and they have both polar and non-polar components.
- In the body (and other aqueous environments), these anions form ball-shaped structures called micelles.
- Micelles are able to interact with both polar and non-polar molecules, and serve as soaps/detergents.

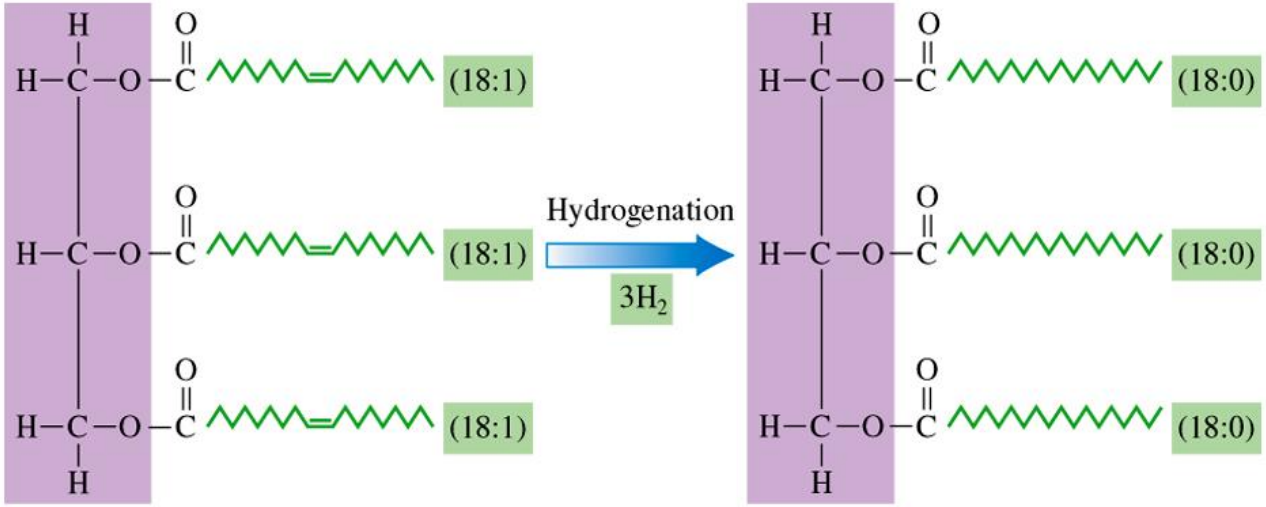




Chemical reactions of triacylglycerols

Reactions of triacylglycerols

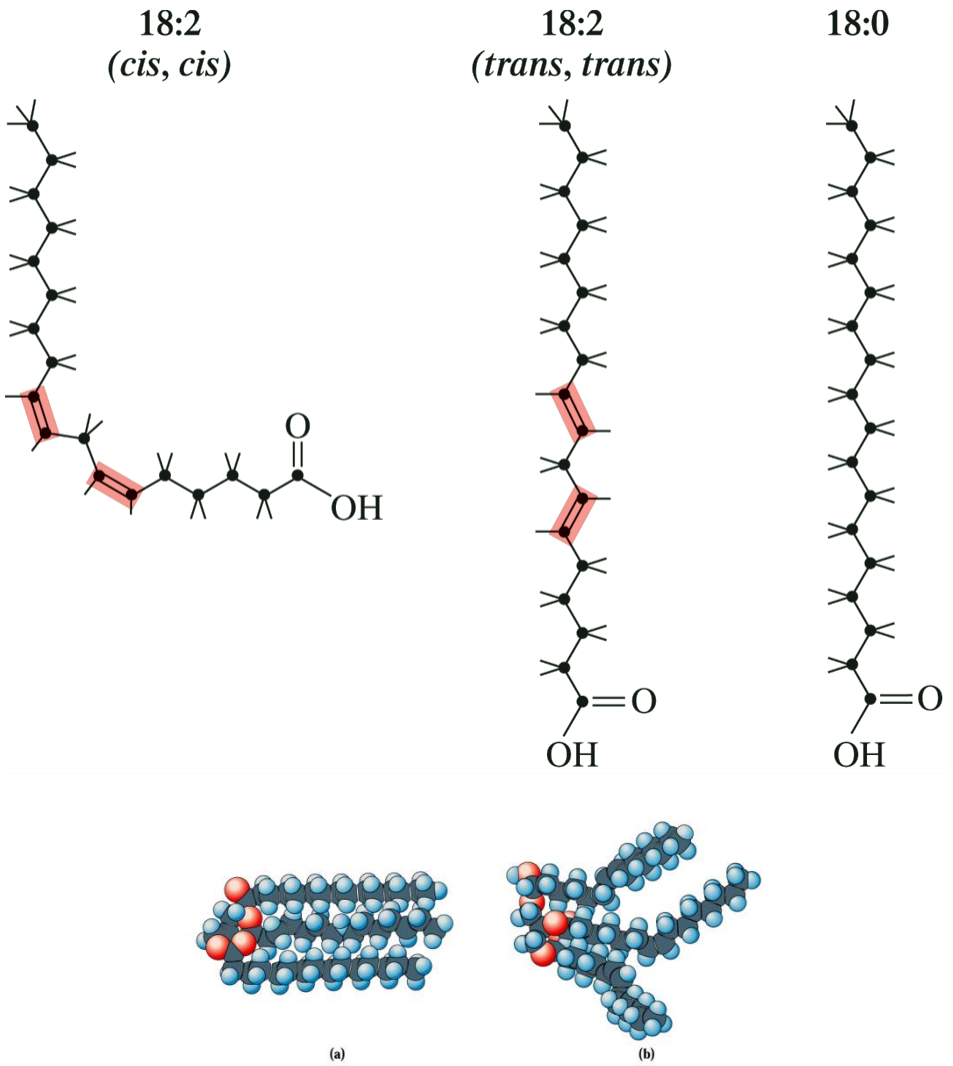
- **Hydrogenation** of a triacylglycerol can happen if the triacylglycerol has unsaturated fatty acid chains.
- Hydrogenation of the double bonds (adding H_2 across the C-C double bond) converts C-C double bonds to C-C single bonds.



Chemical reactions of triacylglycerols

Reactions of triacylglycerols

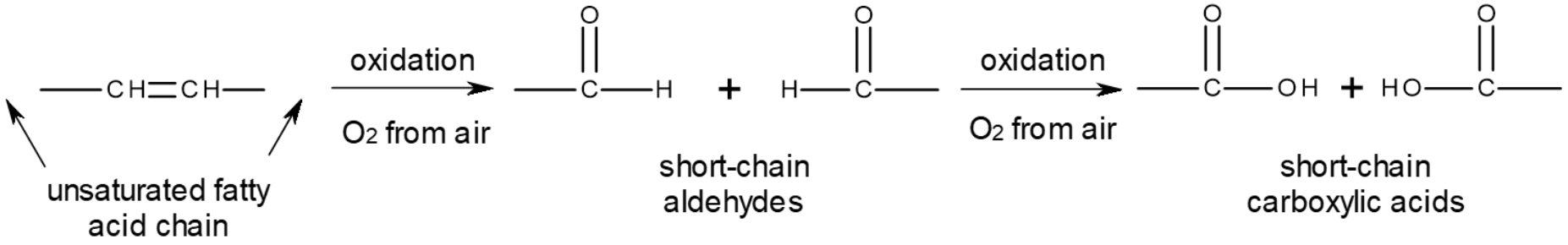
- When hydrogenation occurs, the physical properties of a triacylglycerol will change. For example, more saturated chains will lower the melting point of the triacylglycerol, perhaps causing it to become a semisolid/solid at room temperature.
- This reaction is used in the food industry to make margarine, peanut butter, and other partially hydrogenated products; however, partial hydrogenation also results in the conversion of *cis*-double bonds to *trans*-double bonds (trans-fats).



Chemical reactions of triacylglycerols

Reactions of triacylglycerols

- **Oxidation** of the C-C double bonds can also occur.
- In this process, both bonds in the double bond are broken and two C=O groups are made:



- This kind of reaction occurs when fats become rancid (decompose in air), producing compounds that have bad smells (short-chain aldehydes and carboxylic acids).

CHEMICAL PROPERTIES

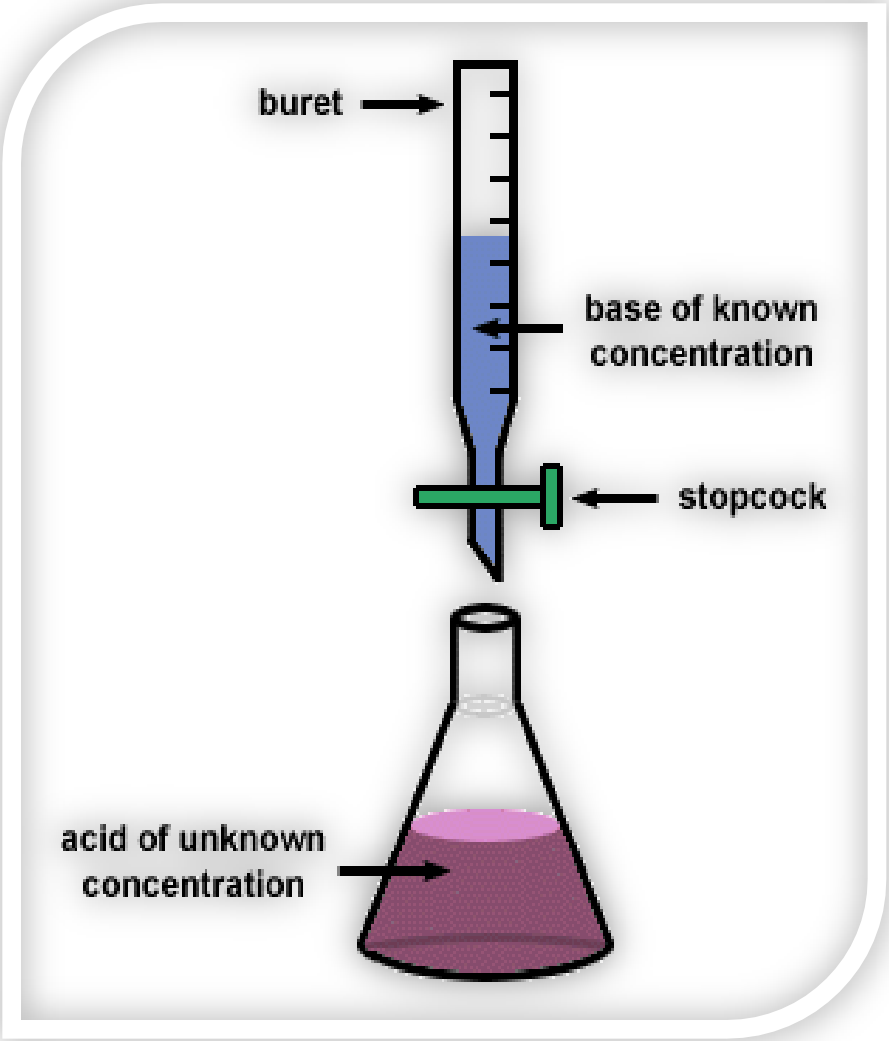
i. Acid Values
ii. Saponification Value
iii. Ester Value
iv. Iodine Value
v. Unsaponifiable Matter
vi. Acetyl value

CHEMICAL PROPERTIES

ACID VALUE

- Acid value: Refers to the number of mg of KOH needed to neutralize the free acids in 1 g of oil.

- High acid values occur in oils which are rancid.



CHEMICAL PROPERTIES

SAPONIFICATION VALUE

- The hydrolysis reaction of lipids (with KOH above) can be used to determine the saponification value of the oil.
- Saponification value is expressed as the number of mg of KOH needed to neutralize the free acids in, and to hydrolyse the esters in, 1 g of the substance (oil).



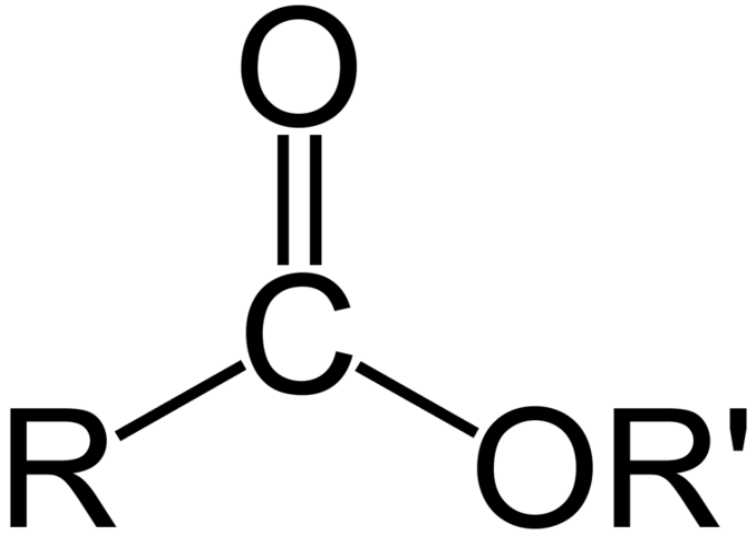
CHEMICAL PROPERTIES

- gives some clue as to the average size of fatty acids in a given sample of fat
  - butter (large proportion of short chain FAs) sap. no. 220 – 230
  - oleomargarine (long chain FAs) sap. No is 195 or less

CHEMICAL PROPERTIES

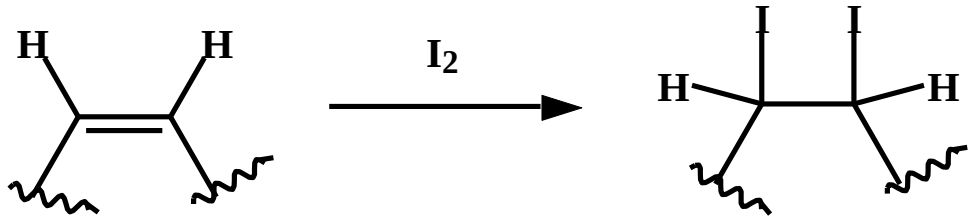
ESTER VALUE

- Ester value: The difference between the saponification and acid values.



CHEMICAL PROPERTIES

Iodine number

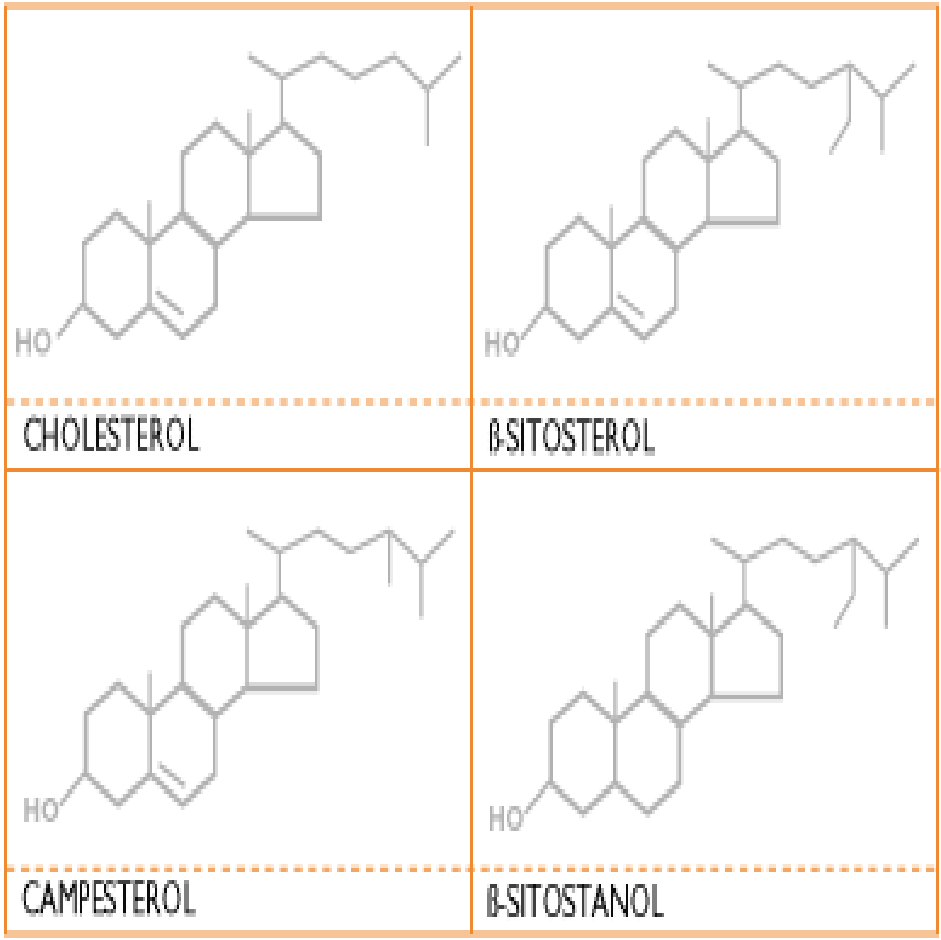


- Measures the degree of unsaturation in a given amount of fat or oil
- The iodine number is the number of grams of iodine absorbed by 100 grams of fat
  - Cottonseed oil: 103 –111
  - Olive oil: 79 – 88
  - Linseed oil: 175 –202
- Frequently used to determine adulteration of commercial lots of oils
- Oils which partially resinify on exposure to air are known as drying oils.
- These oils have high iodine values.
  - E.g. linseed oil

CHEMICAL PROPERTIES

UNSAPONIFIABLE MATTER

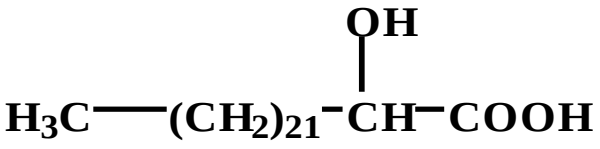
- Unsaponifiable matter consits of compounds such as sterols, which remains after saponification of the triglycerides and the removal of the glycerol and soaps (by using solvents).



CHEMICAL PROPERTIES

Acetyl number

- Some fatty acids have hydroxyl groups

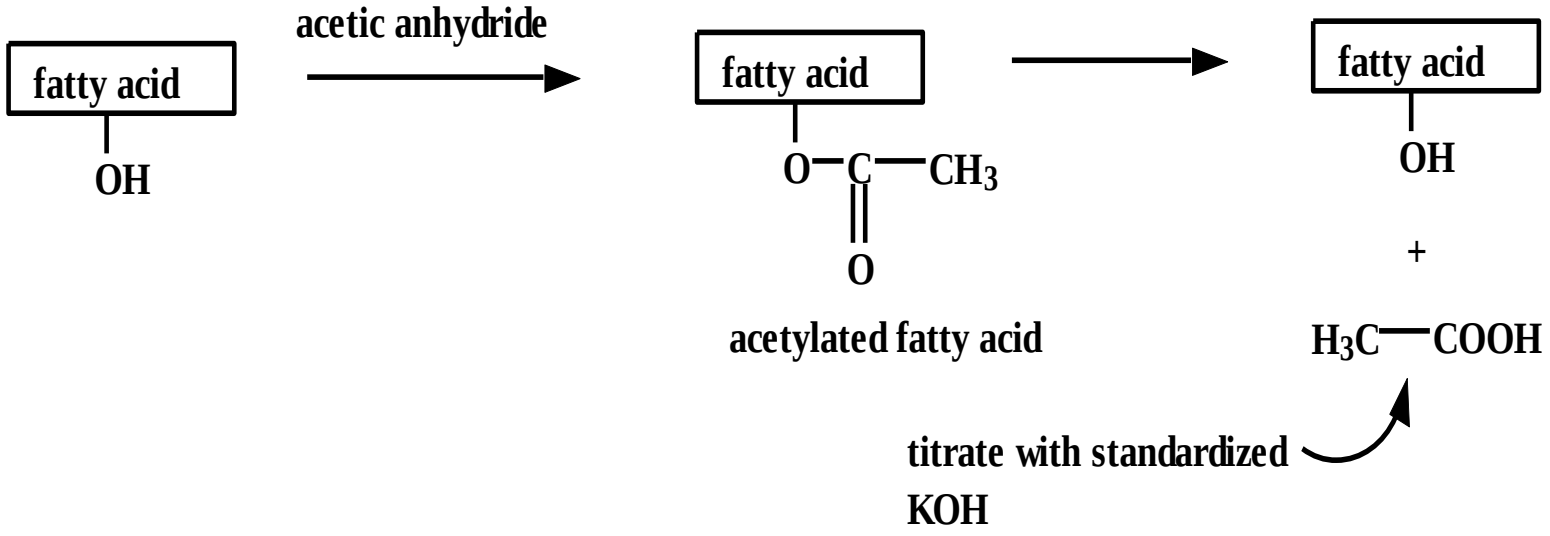


cerebronic acid



ricinoleic acid

The acetyl number gives the proportion of these hydroxyl-containing fatty acids in a given sample of fat or oil



CHEMICAL PROPERTIES

- the acetyl number is the number of milligrams of KOH needed to neutralize the acetic acid of 1 Gm of acetylated fat
- The oil is first acetylated with acetic anhydride, which combines with any hydroxyl group present.
  - examples:
    - castor oil – 146 –150
    - cod liver oil – 1.1
    - cottonseed oil – 21 – 25
    - olive oil – 10.5
    - peanut oil – 3.5

CHEMICAL PROPERTIES

- Because these are absent from most fatty acids, the small acetyl values usually obtained are due to relatively small amounts of sterols.
- But: In an oil such as castor oil, the acetyl value is high (146-150), due to the large amounts of the hydroxyl acid – ricinoleic acid.

CHEMICAL PROPERTIES

Reichert – Meissl number

- Measures the amount of volatile fatty acids (low MW and water soluble Fats)
- The R-M number is the number of milliliters of 0.1N alkali required to neutralize the soluble fatty acids distilled from 5 Gm of fat.

Uses

- Readily absorbed through skin
  - ointments
- Protect from entry of water
  - eg cod liver oil, linseed oil
- Vehicles for injections
- Waxes in enteric coatings
- Pharmacological substances
  - vitamin A and D in cod liver oil
- Food source
  - eg peanut oil
  - may contain essential fatty acids (linoleic acid, arachidonic acid required in prostaglandin formation)

Production of fixed oils and fats

- Fixed oils and fats of vegetable origin are obtained by:

1. Extraction by expression
   - Cold Expression
     - Oil is normally less viscous with less odour (better quality but low yield)
   - Hot Expression
     - is used only for technical oils (not medicinal oils).
2. Extraction with solvents

Production of fixed oils and fats

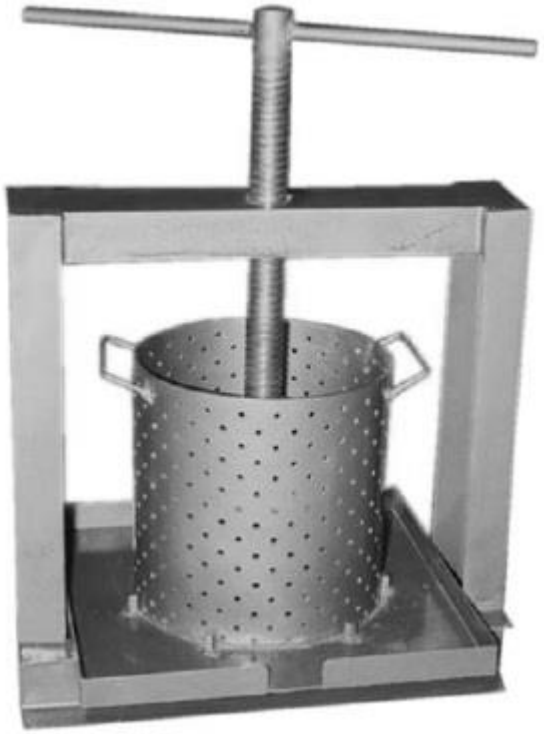
- Fixed oils and fats of vegetable origin are obtained by:

  1. Extraction by expression
     - A. Cold expression
     - B. Hot expression
- Fixed oils are obtained by expression in hydraulic presses. If the expression is carried out in the cold, the oil is known as a virgin oil or a cold-pressed oil, In contrast, if the expression is carried out in heat, the oil is known as a hot-pressed oil.

1. Extraction by Expression

- Screw presses are normally used to express oils from plant material because they give a better yield than older hydraulic presses.
  - They also operate at higher pressure and continuously(not in batches)

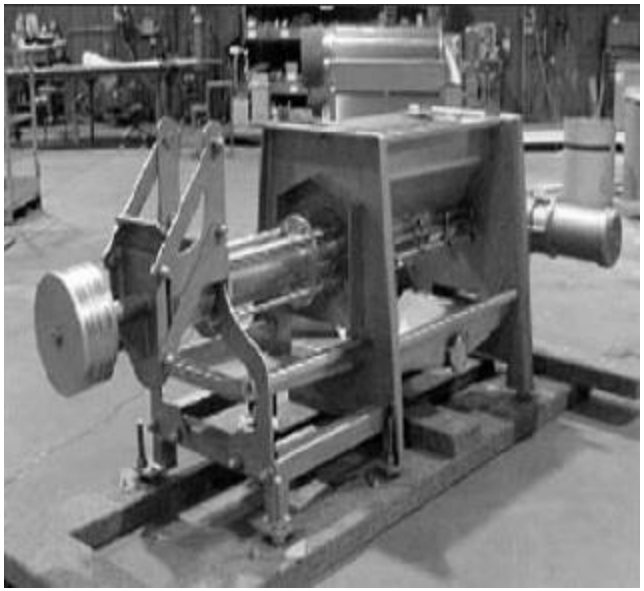
- Before undergoing expression, seeds rich in proteins are cooked at 90°C frees the oil by bursting the cell structures and coagulates the proteins. A fast drying step normally follows



A. Cold Expression

- Oils for medical uses are extracted at room temperature. Only a portion of the oil is obtained.
- Oil is normally less viscous with less odour than with hot expression(better quality but low yield).

- The remaining pressed material may then be ground, heated and pressed to express any remaining oil.
- The pressed material has nutritive value and is used as cattle feed.



B. Hot Expression

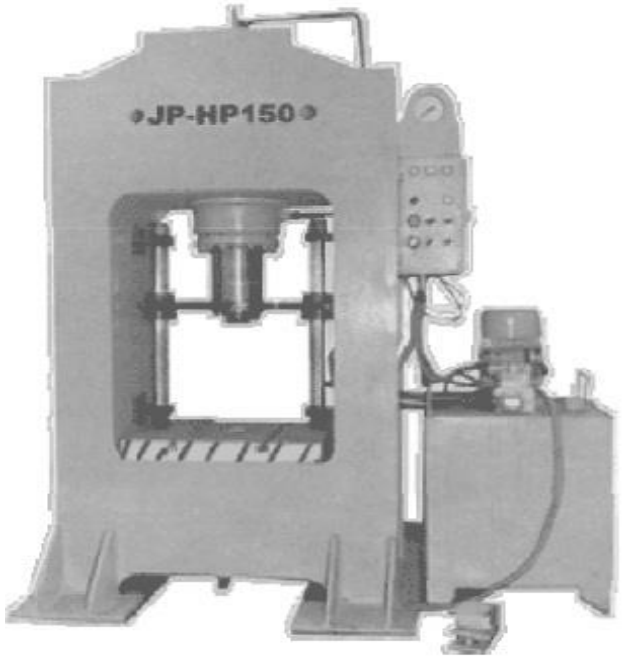
- The residue left after cold expression is broken down and treated with steam.
- This causes the remaining oil cells to rupture.
- Even after hot expression, 10% of the oil remains in the plant material.
- Types of machines used for Expression

  - i. Hydraulic Press
  - ii. Oil Expeller.

- i. **Hydraulic press**
  - Series of corrugated plates between which the plant material is placed.
  - As the ram rises, the P forces the oil out and is collected in a gutter around the ram.
  - The plates are deeply corrugated to provide channels for the oil to escape, but also to prevent the spreading of the material during pressing.

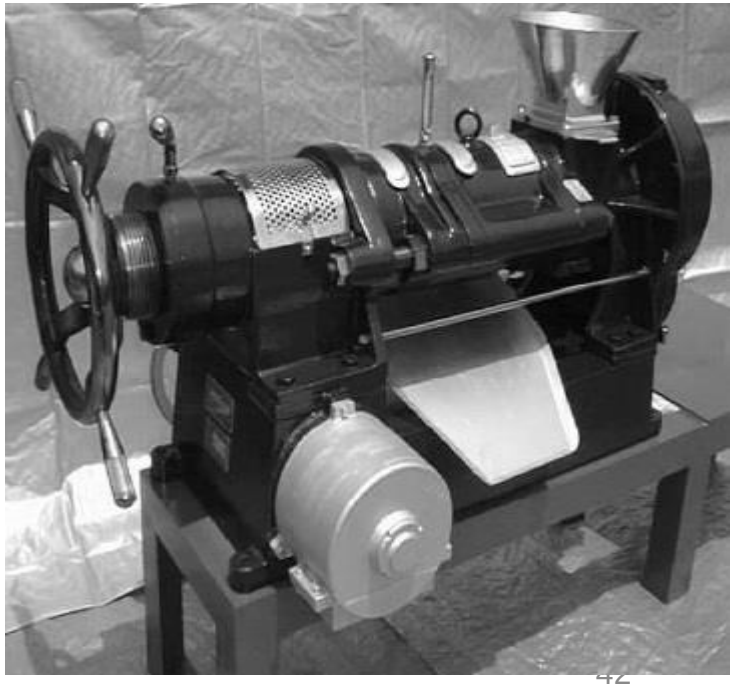
◦ This type of press is used for seeds not containing a very high amount of oil.

◦ Disadvantage: This method is not continuous (plates have to be cleaned after every pressing).



- **ii. Oil Expeller**
  - The material is spread continuously through a perforated steel cylinder through which it is forced by a spiral screw.
  - The pressed material then escapes at the opposite end through a choking cone.

  - Advantages: Continuous process;<manual labour.
  - Disadvantage: Suitable only for seeds with a high amount of oil.



2. Extraction by solvents

- Sometimes organic solvents are used for the extraction of oils. This type of method is used only for technical oils(not medicinal oils).
- Seeds used: Intact or partially extracted by expression.
- Animal fats are separated from other tissues by rendering with steam, with or without pressure. The heat melts the fat, which rises to the top and may be separated by decantation.

REFINING OF CRUDE OIL

- Crude oil may contain water, FFAs, lecithins, resins, pigments (carotenes, chlorophyll), sterols, waxes, substances with odours & tastes, and external contaminants (pesticides).

Refining consists of the following processes

i. Degumming
   - Mucilage removal

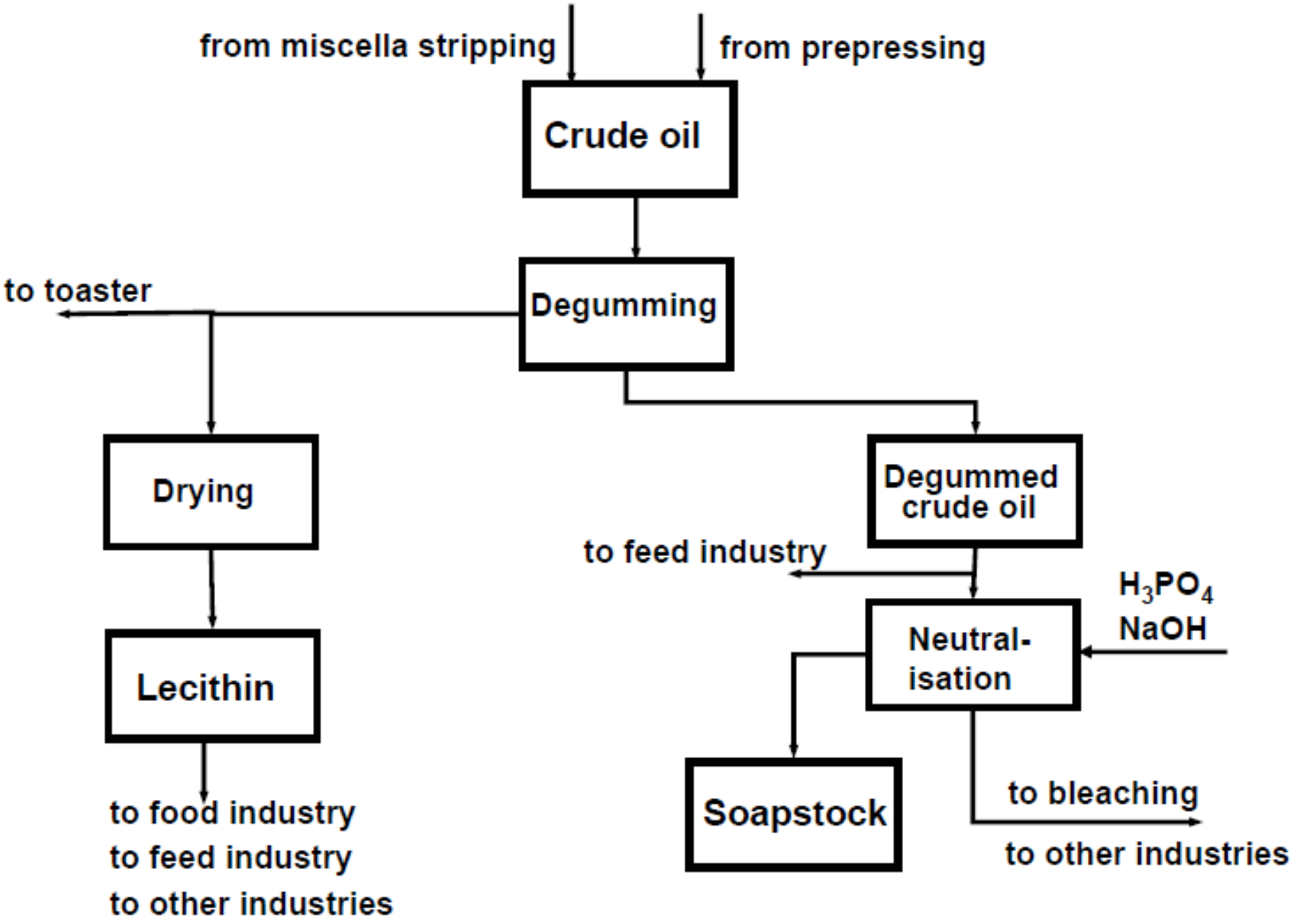
ii. Neutralization
   - FFAs by dilute NaOH.

iii. Bleaching
   - Oil is passed through diatomaceous earths or activated charcoal

iv. Wax removal
   - When cooled (frozen), waxes solidify.

v. Deodorizing
   - eliminated by injecting steam into the very hot oil (>200ºC) under high vacuum of aldehydes and ketons .

Flow Chart chemical Refining



i. Degumming

- Degumming means Mucilage removal
- For ex. To remove lecithins, proteins and other constituents present in the oil in colloidal suspension.
- Method: Hot oil is hydrated → colloids form a dense gel which separates from the lighter oil.
- The gel is discarded and the oil is dried under vacuum
- In most cases, this treatment is replaced by an injection of phosphoric acid into the hot oil
- Phospholipids then precipitate when neutralized by NaOH.

ii. Neutralization

- Fatty acids always present in crude oil, are neutralised by dilute NaOH.
- The soaps formed (soap stock) adsorbs part of the impurities: colouring matter, phenols, sterols, wax esters, traces of metals and miscellaneous oxidation products.
- Excess soap and NaOH are removed by washing with hot water.

iii. Bleaching

- Oil is passed through diatomaceous earths or activated charcoal.
- The bleaching agent (charcoal/earth) is then removed by filtration.

iv. Wax removal

- Crude oil is rich in waxes(sunflower, corn and cotton seed oils).
- When cooled(forzen), waxes solidify.
- The crystallized waxes are then removed by filtration.

v. Deodorising

- Aldehydes & ketones are responsible for unpleasant odours of crude oils.
- These are eliminated by injecting steam into the very hot oil(200^0C) under high vacuum.

OFFICIAL FIXED OILS /FATS

- **Arachis oil**
- **Castor oil**
- **Almond oil**
- **Sesame oil**
- **Theobroma oil**
- **Cod liver oil**
- **Olive oil**
- **Linseed oil**
- **Cotton seed oil**

Arachis oil

(groundnut)

- **Botanical Source**:
  - It is a fixed oil expressed from seeds of *Arachis hypogaea* Linn.

- **Family:** Leguminosae

- **Geographical source**:
  - cultivated in tropical Africa, India, Brazil, China, southern USA and Australia

- **Description**:

Colour	-	Pale yellow coloured liquid
Odour	-	Faint and characteristic
Taste	-	Bland and nut-like
Solubility	-	Slightly soluble in alcohol and soluble in solvent ether, chloroform, and light petroleum(40-60%).

- **Chemical Composition**:
  - Arachis oil consists of glycerides of fatty acids, chiefly
  - **oleic acid** $[CH_3(CH_2)_7CH=CH(CH_2)_7COOH]$ ~ 60%
  - **linoleic acid** $[CH_3(CH_2)_4CH=CH(CH_2CH=CH(CH_2)_7COOH]$ 24%
  - **arachidic acid**$[CH_3(CH_2)_{18}COOH]$
  - **Palmitic acid** $[CH_3(CH_2)_{14}COOH]$

- acid and saponification values similar to olive oil
- **Uses**:
  - Arachis oil is a solvent for intramuscular injections.
  - It is used in preparation of liniments, plasters and soaps.
  - Being a non-drying oil, it is valuable as a lubricant.
  - It produces firm and excellent white soap.
  - Principally it is used as an edible oil.

Castor oil

- **Botanical source**:
  - Castor oil is the fixed oil obtained by cold expression of the seeds of *Ricinus communis*
- **Family**: Euphorbiaceae
- **Geographical source**:
  - Tropical and subtropical countries like India, Africa, Europe, Brazil, Russia, Thailand.
- **Description**:

Colour	-	Pale yellow or colourless liquid
Odour	-	Slight and characteristic
Taste	-	First it is bland but afterwards slightly acrid, and usually nauseatic.
Solubility	-	Soluble in alcohol, miscible in chloroform, solvent ether, glacial acetic acid & petroleum ether

- **Chemical constituents**:
  - Castor oil chiefly contains triglyceride of ricinoleic acid(about 80%).
  - Other glycerides are also present in the drug, where the fatty acids are represented by isoricinoleic, linoleic, stearic and isostrearic acids. The viscosity of castor oil is due to ricinoleic acid.
  - Ricinoleic acid - $CH_3(CH_2)_6CHOHCH_2CH=CH(CH_2)_7COOH$
- **Uses**:
  - Used as cathartic(due to irritant action of ricinoleic acid).
  - Commercially as lubricant.
  - It is used in abortificient paste and ricinoleic acid is used in contraceptive creams and jellies.
  - It is also employed as an emolient in preparation of lip-sticks.
  - As sulphorecinolate in tooth formulation being strong bactericide.

Sesame Oil

- **Biological source**:
  - It is fixed oil obtained by expression from the seeds of *Sesamum indicum* Linn.
- **Family**: Pedaliaceae.
- **Geographical source**:
  - Indigenous to india, and is cultivated in Caribbean islands, china, Japan, Africa and US.
- **Description**:

Colour	-	Pale yellowish liquid
Odour	-	Slight, characteristic
Taste	-	Bland
Solubility	-	Soluble in alcohol, miscible with chloroform, solvent ether, light petroleum and carbon disulphide.

- **Chemical Constituents**:
  - Gingelly oil contians glycerides of higher fatty acids mainly Palmitic , Stearic , Oleic, Linoleic, Linolenic and arachidic acids.
- **Uses:**
  - It is nutritive, demulcent and has got emollient properties.
  - It is used in the preparation of liniments, plasters, ointments and soaps, similar to olive oil.
  - Pharmaceutically, it is used as a vehicle for intramuscular oily injections.
  - After burning, sesame oil yields high quality black ink.
  - **Sesame oil** (also known as **gingelly oil** or **til oil**) is an edible vegetable oil derived from sesame seeds.
  - Besides being used as a cooking oil in South India, it is often used as a flavor enhancer in Chinese, Korean, and to a lesser extent Southeast Asian cuisine.

Theobroma oil

- Cocoa butter
- Obtained from seeds of *Theobroma cacao* (Sterculaceaea)
- Central America, also cultivated in Brazil, W.Africa (Nigeria)
- Solid oil
  - high steric and palmitic acid content (35%, 25%), oleic acid 3%
  - GS_2U 52%
  - melting point 31-35^{0}C
  - low iodine value because saturated
  - most expensive commercial fixed oil (adulterated)
- Mainly used in suppositories

Cod liver oil

- **Biological source**:
  - Cod liver oil is processed from fresh liver of cod fish, *Gadus morrhua* and other species of Gadus
- **Family**: Gadidae
- **Description**:

Colour	-	Pale yellow
Odour	-	Slightly Fishy
Taste	-	Slightly Fishy
Solubility	-	It is freely soluble in chloroform, ether, carbon disulphide, petroleum ether, and slightly soluble in alcohol.

Cod liver oil

- **Chemical constituents**:
  - The medicinal value of oil is due to vitamin A and vitamin D group.
  - About 1g of oil contains not less than 255mcg of vitamin A and 21.25mcg of vitamin D.
  - The oil contains glyceryl esters of oleic, linoelic, gadoleic, myristic, palmitic and other acids.
  - It contains mixed triglycerides, mainly unsaturated C_{16-22} acids and 7% decahexanoic acid
- **Uses**:
  - For Vitamin A and D content (halibut > cod)
  - The oil is used as a nutritive and in treatment of rickets and Tuberculosis.

Olive oil

- salad oil, sweet oil
- **Botanical source**:
  - It is fixed oil expressed from the ripe fruits of *Oleo europoea* Linn.
- **Family**: Oleaceae
- **Geographical source**:
  - Native to palestine and countries near to Mediterranean sea and also in US and Australia.
  - Known in Egypt in 7th century B.C., introduced into Spain early on.
- Description:

Colour	-	Pale yellow or greenish yellow
Odour	-	Slight and characteristic
Taste	-	Bland, faintly acrid
Solubility	-	Slightly soluble in alcohol and miscible with Carbon disulphide, Chloroform and ether.

  - Goes “pasty”/cloudy at 10°C

- **Chemical constituents**:
  - Olive oil contains the triglycerides mainly in the form of olein, palmitin and linolein.
- high iodine value, low acid value

Acids		Glycerides	
Palmitic	12%	GSU_2	60%
Oleic	80%	GU_3	40%
Linoleic	8%		

- **Uses**:
  - Externally it is emollient and soothing agent for inflammaed surfaces.
  - It is used to soften the skin and crusts in eczema and psoriasis.
  - It is used as an ingredient of ear wax. Used in soaps, plasters.
  - Internally it is used as nutrient, demulcent and also as mild laxative.
  - It is used as a vehicle for oily suspension for injection.

Cottonseed oil

- **Biological source:**
  - **Cottonseed oil** is a cooking oil extracted from the seeds of cotton plant of various species, mainly *Gossypium hirsutum* and *Gossypium herbaceum*.
- Cotton grown for oil extraction is one of the big four genetically modified crops grown around the world, next to soy, corn, and rapeseed (canola).
- **Chemical composition:**
  - Its fatty acid profile generally consists of 70% unsaturated fatty acids including 18% monounsaturated (oleic), 52% polyunsaturated (linoleic) and 26% saturated (primarily palmitic and stearic).

Almond oil

- **Biological source**:
  - Obtained from *Prunus amygdalus v. amara* (bitter – used in pharmacy), v. *dulcis* (sweet)

- **Geographical source**:
  - Native to Far East, grown in Mediterranean, N.California

- **Chemical constituents**:
  - Oil is highly unsaturated
    - with oleic acid 77%, linoleic acid 17%
    - Bitter almond oil also contains amygdalin (glycoside) which decomposes to benazldehyde + HCN

- **Uses**:
  - Used in pharmacy in oily injections and ointments
  - prone to oxidation so has to be kept air-free or goes rancid – transfer to smaller bottle to exclude air

Linseed oil

- **Synonyms:** flax seed oil
- **Biological source**:
  - It consists of fixed oil obtained from the dried fully ripe seeds of the palnt *Linum usitatissimum* Linn.
  - **Family**: Linaceae.
  - The oil is obtained by cold pressing, sometimes followed by solvent extraction.
- **Macroscopic characters**:
  - Colour - Pale yellow coloured clear liquid
  - Odour - Characteristic
  - Taste - Pleasant
  - Solubility - Soluble in alcohol, insoluble in water and miscible with ether, petroleum ether and chloroform.

- **Chemical constituents**:
  - It contains the glycerides of palmitic, stearic, oleic, linoleic and linolenic acids.
  - The triglycerides are primarily derived from the following fatty acids:
  - The saturated acids palmitic acid (about 7%) and stearic acid (3.4-4.6%),
  - The monounsaturated oleic acid (18.5-22.6%),
  - The doubly unsaturated linoleic acid (14.2-17%),
  - The triply unsaturated omega-3 fatty acid α-linolenic acid (51.9-55.2%).
  - The unsaponifiable matter of the oil contains considerable quantity of sterols, tocopherols and sqaulene.

- **Uses**:
  - Linseed oil is a so-called drying oil, which means that it hardens upon exposure to air. It is the most important drying oil and hence large quantities are used for paints and varnishes.
  - Medicinally it is mainly recommended for external applications like lotions and liniments.
  - It is used in treatment of scabies and other skin diseases alongwith sulphur.
  - Industrially it is as important oil used for various purposes such as in the manufacture of soaps, grases, polishes, polymers etc.
  - Due to its polymer-like properties linseed oil is used on its own or blended with other oils, resins and solvents as an impregnator and varnish in wood finishing, as a pigment binder in oil paints, as a plasticizer and hardener in putty and in the manufacture of linoleum.

waxes

- Simple esters of fatty acids (usually saturated with long chain monohydric alcohols)
- Waxes are organic compounds that characteristically consist of a long alkyl chains.
- Natural waxes are typically esters of fatty acids and long chain alcohols.
- Synthetic waxes are long-chain hydrocarbons lacking functional groups.
- Waxes contain appreciable quantities of
  - esters derived from higher monohydric alcohols (one –OH group) of the methyl alcohol series
  - combined with fatty acids (C_{16} – C_{32})

waxes

- Most are solids at room temperature
  - Can only be saponified by alcoholic alkali
  - Often contain free acids, hydrocarbons, free alcohols and sterols
  - Saponification and acid values higher, iodine values lower
- commercially important examples
  - Animal: spermaceti, beeswax, ‘wool fat’
  - Vegetable: carnuba

Properties of waxes

- Abundant in plants leaves “cuticle” layer
- Insoluble in alcohol
- Soluble in organic solvents
- Viscous solid substances
- More resistance to saponification
  - Waxes can be saponified but with difficulty
  - Can be only saponified by alcoholic alkali

Types of Waxes

Animal waxes

- **Beeswax** - produced by honey bees
- **Chinese wax** - produced by the scale insect *Ceroplastes ceriferus*
- **Earwax** - found in the human ear.
- **Lanolin (wool wax)** - from the sebaceous glands of sheep
- **Shellac wax** - from the lac insect *Kerria lacca*
- **Spermaceti** - from the head cavities and blubber of the sperm whale

Vegetable waxes

- **Bayberry wax** - from the surface wax of the fruits of the bayberry shrub, *Myrica faya*
- **Candelilla wax** - from the Mexican shrubs *Euphorbia cerifera* and *Euphorbia antisyphilitica*
- **Carnauba wax** - from the leaves of the Carnauba palm, *Copernica cerifera*
- **Castor wax** - catalytically hydrogenated castor oil
- **Esparto wax** - a byproduct of making paper from esparto grass, (*Macrochloa tenacissima*)
- **Japan wax** - a vegetable triglyceride (not a true wax), from the berries of *Rhus* and *Toxicodendron* species
- **Jojoba oil** - a replacement for spermaceti, jojoba is pressed from the seeds of the jojoba bush, *Simmondsia chinensis*
- **Rice bran wax** - obtained from rice bran (*Oryza sativa*)
- **Soy wax** - from soybean oil

Animal Waxes

- ***Beeswax***
  - This wax is an abdominal secretion of bees (*Apis mellifera*), its colour being dependent of the flowers gathered by these insects.
  - Bees wax is easily saponifiable and emulsifiable because of its content in free fatty acids, diols and hydroxyacids
  - Its main components are palmitate, palmitoleate, hydroxypalmitate and oleate esters of long-chain alcohols (C30-32)
  - yellow and white from honeycomb
  - ester:acid ratio value 3.3-4.2
  - Used in paraffin ointment, plasters and enteric coating

Animal Waxes

- **Chinese wax (insect wax)**
  - This wax is secreted by insects (*Coccus ceriferus*) and laid on tree branches (1500 insects are needed to produce 1g chinese wax).
  - That insect is cultivated in China.
  - Besides an important content in esters (about 83%), this wax includes some free acids, alcohols (up to 1%) and hydrocarbons (2 to 3%).
  - Chemically, the esters are formed of chains with 46 up to 60 carbon atoms, the majority of alcohols and acids having 26 or 28 carbon atoms.
  - The purified wax is used to make candles and polish.

Animal Waxes

- ***Wool fat (Lanolin)***
  - This material is secreted by sheep (*Ovis aries*) sebaceous glands and collected from crude wool by dilute alkali or detergent washing.
  - Unwashed wool contains about 10-24% of greasy matter and a small proportion of salts of long-chain fatty acids.
  - Chemical composition:
    - esters of cholesteryl and isocholesteryl + estolidic 32-33%
    - esters of normal aliphatic alcohols with fatty acids 48-49%
  - Used as an emollient base for creams and ointments
    - major component of most ointments
    - melting point 30-42°C
    - readily absorbed through skin
    - absorbs twice its own weight in water so makes an emulsion

Animal Waxes

- **Shellac wax**
  - This wax (known also as lac wax) is produced by a cochineal insect (*Tachardia lacca*) native of India.
  - It contains a majority of fatty esters (70-82%), free fatty alcohols (8-14%), acids (1-4%) and hydrocarbons (1-6%).
  - The esters are formed of chains of 28 up to 34 carbon atoms.
  - This wax is used in varnish industry and may replace carnauba wax.

Animal Waxes

- ***Spermaceti***
  - Spermaceti is extracted by cooling (11% of the initial oil) from adipose tissues and also collected from a big cavity in the head of a cachalot (*Physeter macrocephalus*) known as sperm whale and bottle-nosed whales (*Hyperodon rostratus*) just above right nostril) – 500lb from 1 whale
  - Sonar part of it contains fatty esters (65-95%) but also triglycerides (5-30%), free alcohols (1-5%) and acids (0-3%).
  - Simple esters of cetyl alcohol $CH_3(CH_2)_4CH_2OH$ – cetyl palmitate, cetyl myistate 90-93%
  - Its melting point is 42-50°C.
  - No longer used pharmaceutically, can be replaced by jojoba oil

Vegetable waxes

- **Carnauba wax (**queen of waxes)
  - This wax is secreted from the cuticle of leaves of a Brasilian palm tree (*Copernicia prunifera cerifera*), about 100 g for one tree in a year.
  - It contains mainly fatty esters (80-85%), free alcohols (10-15%), acids (3-6%) and hydrocarbons (1-3%).an adulterate of beeswax
  - This wax is the hardest and highest melting of the natural waxes (melting point : 78-85°C) and is used mainly mixed to bee wax to make various polishes for shoes, floor and furniture but also in cosmetics (lipsticks, creams) and in food industry (glazes for candies, gums, fruit coatings ...).
  - It is used also in the paper industry for paper coating
  - Used in tablet coatings.

Vegetable waxes

➢ Candelilla wax

- This wax is produced by small shrubs from Mexico, *Euphorbia cerifera* and *E. antisyphilitica* (*Euphorbiaceae*).
- The wax is extracted by boiling the plant.
- It contains hydrocarbons (about 50%), esters (28-29%), alcohols, free fatty acids (7-9%), and resins (12-14% triterpenoid esters). Its melting point is 67-79°C.
- This wax is used in cosmetics (lip balms and lotion bars), pharmaceutics and in food stuffs to improve stability and texture as a substitute to beeswax (melting point : 66-71°C).
- One of candelilla's major outlets was a binder for chewing gums.

Vegetable waxes

➢ Rice bran oil

- Rice bran from the milling of rice, Oryza sativa, contains a wax mixed with triglycerides.
- The melting point of the pure wax is 75-80°C.
- It contains esters of fatty acids (26 to 30 carbon atoms) and long-chain alcohols (C26 to C30) and a large amount of unsaponifiable matter (55-67%).
- That wax is much used as a constituent of chocolate enrobers, various fruit and vegetable coating and as a lipstick.

ADULTERATION OF FIXED OILS

- Cheap oils are often mixed with more expensive oils as a form of adulteration.
- 3 of the most commonly used to adulterate are
  - Sesame oil (detected by Baudouin's test).
  - Cotton seed oil (Halphen's test)
  - Arachis oil (Bellier's test)

Why are lipids biologically significant?

- They make up cell membranes.
- They make up waterproof coverings.
- They can be used to store energy. (More than twice the energy holding capacity of a carbohydrate…)
- They are ‘chemical messengers.’ (Steroids)

Applications of fixed oils and fats

1. Soap manufacture
2. Suppositories, tablet coating
3. Dietary supplements
4. Emulsifying agents
5. Manufacture of paints, varnishes and lubricants
6. Therapeutic uses (castor oil).

USES OF WAX

1. Wax is used in pharmacy to make soft ointments harder and to prepare lip salves.
2. The technical uses of waxes are substantial, e.g. in shoe polishes and car waxes.
3. Waxes are mainly consumed industrially as components of complex formulations, often for coatings.
4. The main use of polyethylene and polypropylene waxes is in the formulation of colourants for plastics.
5. Waxes confer matting effects and wear resistance to paints.
6. Polyethylene waxes are incorporated into inks in the form of dispersions to decrease friction.
7. They are employed as release agents.
8. They are also used as slip agents, e.g. in furniture, and corrosion resistance.